

Question	Answer	Mark	Mark scheme	Additional guidance
19	$-\frac{1}{2} \pm \frac{1}{2}\sqrt{2}$	P1	for using a common denominator, eg $\frac{x+1}{x(x+1)} - \frac{x}{x(x+1)} (=4)$ or $\frac{x+1-x}{x(x+1)} (=4)$ or $x+1-x=4x(x+1)$	Note we don't need to see “= 0”; just the LHS is sufficient Accept other forms of the 3 term quadratic, eg $4x^2 + 4x = 1$ Correct use of formula or completing the square
		P1	for expanding and rearranging to get $4x^2 + 4x - 1 (=0)$	
		P1	(dep P1) ft for a method to solve their 3 term quadratic equation, eg $\frac{-4 \pm \sqrt{4^2 - 4 \times 4 \times -1}}{2 \times 4}$ or $4 \left[\left(x + \frac{1}{2} \right)^2 - \left(\frac{1}{2} \right)^2 \right] - 1 = 0$ oe	
		A1	for values of x , eg $\frac{-4 \pm \sqrt{32}}{8}$ or $\pm \sqrt{\frac{1}{2}} - \frac{1}{2}$ oe	Accept $a = -\frac{1}{2}, b = \frac{1}{2}$ or $a = -\frac{1}{2}, b = -\frac{1}{2}$
		A1	for $-\frac{1}{2} \pm \frac{1}{2}\sqrt{2}$ oe in the form $a \pm b\sqrt{2}$ where a and b are fractions	